Exchange of Limit and Expectation

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References

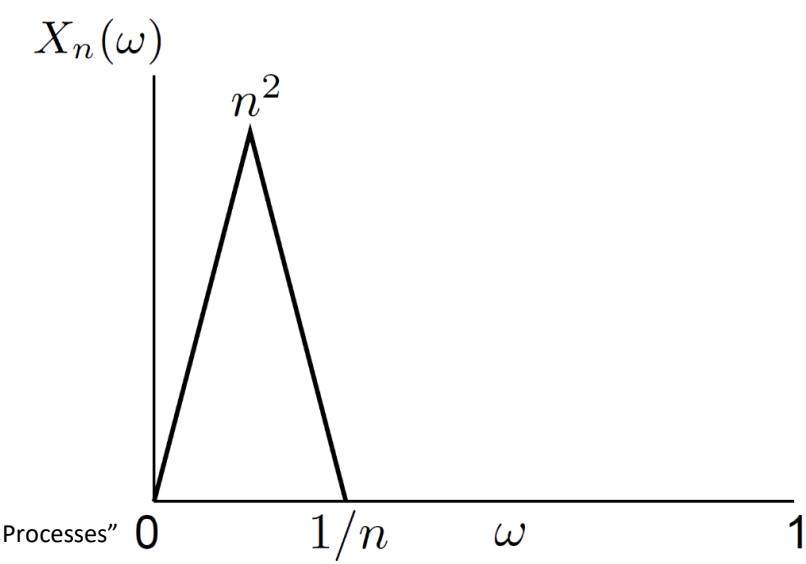
- A. Kumar, “Discrete Event Stochastic Processes”: Sections 1.4.1 and 3.2
- S.M. Ross, “Stochastic Processes”: Chapter 3

Proof of Expression for Renewal Function

- Recall:
  - $N(t)$: number of events that occur up to time t
  - $m(t) = E[N(t)]$
  - **Proposition**: For all $t \geq 0$, $m(t) = \sum_{n=1}^{\infty} F_n(t)$
  - **Proof**:
    - $N(t) = \sum_{n=1}^{\infty} 1_{\{S_n \leq t\}}$
    - Take expectations on both sides
- In this proof, we assumed that we can exchange the expectation with the infinite summation
- In other words, letting $Y_M = \sum_{n=1}^{M} 1_{\{S_n \leq t\}}$, we assumed that the following exchange of limit and expectation is valid:
  - $\lim_{M \to \infty} E(Y_M) = E\left(\lim_{M \to \infty} Y_M\right)$
- Such an exchange is not always valid; an example in which it is not valid is as follows

Example

- Let the sample space $\Omega = [0,1]$, with the uniform distribution over it
- Define r.v.s $X_1, X_2, \ldots$ as shown in Fig.
- X_n is zero over $\left[\frac{1}{n}, 1\right]$
- For each $\omega \in (0,1]$:
  - $\lim_{n\to\infty} X_n(\omega) = 0$
- Let $X(\omega) = 0$ for all $\omega \in \Omega$
- Then $X_n \to X$ w.p. 1
- $E(X_n)$:
  - $\frac{1}{2} \times \frac{1}{n} \times n^2 = \frac{n}{2}$
- So, $\lim_{n\to\infty} E(X_n) = \infty$
- However, $E\left(\lim_{n\to\infty} X_n\right) = E(X) = 0$
- Hence:
  - $\lim_{n\to\infty} E(X_n) \neq E\left(\lim_{n\to\infty} X_n\right)$



Ref: A. Kumar, "Discrete Event Stochastic Processes"

Validity of Exchange of Limit and Expectation

- The following two results provide conditions that ensure validity of exchange of expectation and a.s. limit of r.v.s:
  - ❑Monotone Convergence Theorem (MCT)
  - ❑Dominated Convergence Theorem (DCT)

Monotone Convergence Theorem

- **Theorem**:
  - Suppose $X_1, X_2, \ldots$ is a sequence of r.v.s that converges a.s. to the r.v. X and the following two conditions hold:
    - *a)* X_n is non-negative for every $n \geq 1$, and
    - *b)* $X_1 \leq X_2 \leq \cdots \leq X_n \ldots$ (i.e., the sequence of r.v.s is monotonically non-decreasing),
  - Then:
    - $\lim_{n \to \infty} E(X_n) = E(X)$
- **Note**: In the situation of above theorem, it is possible that $E(X) = \infty$

Proof of Expression for Renewal Function: Justification Using MCT

- Recall:
  - **Proposition**: For all $t \geq 0$, $m(t) = \sum_{n=1}^{\infty} F_n(t)$
  - **Proof**:
    - $N(t) = \sum_{n=1}^{\infty} 1_{\{S_n \leq t\}}$
    - Take expectations on both sides
  - Letting $Y_M = \sum_{n=1}^{M} 1_{\{S_n \leq t\}}$, we need to prove that the following exchange of limit and expectation is valid:

    1) $\lim_{M \to \infty} E(Y_M) = E\left(\lim_{M \to \infty} Y_M\right)$
- Note that $Y_M \geq 0, \forall M$ and $Y_1 \leq Y_2 \leq \cdots \leq Y_M \ldots$
- So, 1) follows from MCT

Proof of Wald's Equation

- Recall**:**
  - **Theorem**:
    - Suppose:
      - $X_1, X_2, \ldots$ is a sequence of i.i.d. r.v.s such that $E[|X_n|] < \infty$
      - N is a stopping time for this sequence s.t. $E[N] < \infty$
    - Then:
      - $E[\sum_{n=1}^{N} X_n] = E[N]E[X]$
  - **Proof**:
    - $S_N = \sum_{n=1}^{N} X_n$:
      - $\sum_{n=1}^{\infty} X_n \, 1_{\{N \geq n\}}$
    - So, $E[S_N]$:
      - *1)* $\sum_{n=1}^{\infty} E\left[X_n 1_{\{N \geq n\}}\right]$
      - $\sum_{n=1}^{\infty} E[X_n] \, E\left[1_{\{N \geq n\}}\right]$ since the event $\{N \geq n\}$ is independent of X_n
      - $E[X](\sum_{n=1}^{\infty} P\{N \geq n\})$
      - $E[X] \, E[N]$
- Let $Z_M = \sum_{n=1}^{M} X_n \, 1_{\{N \geq n\}}$; to obtain step 1), we assumed that:

  2) $\lim_{M \to \infty} E(Z_M) = E\left(\lim_{M \to \infty} Z_M\right)$
- Note that 2) does not follow from MCT

Dominated Convergence Theorem

- **Theorem**:
  - ❑ Suppose $X_1, X_2, \ldots$ is a sequence of r.v.s that converges a.s. to the r.v. X and the following two conditions hold:
    - a) There is a r.v. Y such that $|X_n| \leq Y, \forall n \geq 1$
      - ○ i.e., the sequence X_n, $n \geq 1$, is dominated by the r.v. Y
    - *b)* $E(Y) < \infty$.
  - ❑ Then:
    - ○ $E(X) < \infty$ and
    - ○ $\lim_{n \to \infty} E(X_n) = E(X)$
- **Note**: When Y is a finite constant in the DCT, then the result is called the *bounded convergence theorem*

Proof of Wald's Equation: Justification Using DCT

- Recall:
  - **Theorem**:
    - Suppose:
      - $X_1, X_2, \ldots$ is a sequence of i.i.d. r.v.s such that $E[|X_n|] < \infty$
      - N is a stopping time for this sequence s.t. $E[N] < \infty$
    - Then:
      - $E[\sum_{n=1}^{N} X_n] = E[N]E[X]$
  - **Proof**:
    - $S_N = \sum_{n=1}^{N} X_n$:
      - $\sum_{n=1}^{\infty} X_n \, 1_{\{N \geq n\}}$
    - So, $E[S_N]$:
      - *1)* $\sum_{n=1}^{\infty} E\left[X_n 1_{\{N \geq n\}}\right]$
      - $\sum_{n=1}^{\infty} E[X_n] \, E\left[1_{\{N \geq n\}}\right]$ since the event $\{N \geq n\}$ is independent of X_n
      - $E[X](\sum_{n=1}^{\infty} P\{N \geq n\})$
      - $E[X] \, E[N]$
- Let $Z_M = \sum_{n=1}^{M} X_n \, 1_{\{N \geq n\}}$; to obtain step 1), we assumed that:

  2) $\lim_{M \to \infty} E(Z_M) = E\left(\lim_{M \to \infty} Z_M\right)$
- To justify 2), let $Z = \sum_{n=1}^{\infty} |X_n| \, 1_{\{N \geq n\}}$
- Then by the MCT and the facts that $E[|X_n|] < \infty$, $E[N] < \infty$, and N is a stopping time for $\{X_n, n \geq 1\}$, it follows that $E(Z) < \infty$
- Also, $|Z_M| \leq Z$ for all $M \geq 1$; so, 2) follows from the DCT

CDF of $S_{N(t)}$: Justification Using MCT

- Recall:
  - For all $x \geq 0$, $m(x) = \sum_{n=1}^{\infty} F_n(x)$
  - **Lemma**: For all t and s such that $0 \leq s \leq t$:
    - $P\{S_{N(t)} \leq s\} = F^c(t) + \int_0^s F^c(t-y)dm(y)$
  - **Proof**: $P\{S_{N(t)} \leq s\}$:
    - $\sum_{n=0}^{\infty} P\{S_n \leq s, S_{n+1} > t\}$
    - $F^c(t) + \sum_{n=1}^{\infty} \int_0^{\infty} P\{S_n \leq s, S_{n+1} > t | S_n = y\} dF_n(y)$
    - $F^c(t) + \sum_{n=1}^{\infty} \int_0^s F^c(t-y) dF_n(y)$
    - *1)* $F^c(t) + \int_0^s F^c(t-y) d(\sum_{n=1}^{\infty} F_n(y))$
- Exchange of infinite summation and integral to obtain 1) is valid because of:
  - the fact that all terms are non-negative and the MCT for functions, which states the following:
    - If $0 \leq f_1(x) \leq f_2(x) \leq \cdots$ and $\lim_{n\to\infty} f_n(x) = f(x)$, then $\lim_{n\to\infty} \int f_n(x)dx = \int f(x)dx$